

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in materials characterization, laboratory data quality, root-cause analysis, and DOE-guided process development across polymers, thin films, and functional molecular materials. Experienced translating multi-technique test data into SOP-ready controls, technical recommendations, and reproducible material-quality decisions.

CORE COMPETENCIES

Materials Laboratory Testing

Data Quality & Traceability

Test Method Qualification

Thermal Analysis

Failure Analysis

Calibration & SOPs

DOE/SPC

Technical Training

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, defining material-quality readouts, process windows, scale-up specifications, and SOP-ready controls for reproducible laboratory execution.
- Translate spectroscopy, thermal-analysis, and process data into technical reports, troubleshooting recommendations, and decision criteria for material selection and process improvement.
- Train junior researchers on experimental execution, documentation discipline, and interpretation of characterization data for repeatable results.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Assessed reliability risks in functional electrochromic materials by identifying degradation, delamination, and redox-instability failure modes across 15+ variants using spectroscopy, electrochemistry, thermal analysis, NMR/GPC, and MS.
- Built structure-property-processing datasets that connected purity, stability, and performance to experimental conditions, data anomalies, and material-design recommendations.
- Documented experimental methods and data-quality checks to support reproducibility, root-cause analysis, and technical communication.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded thin-film coating process development, linking CVD/surface preparation to morphology, interface quality, optical performance, and reproducible process controls.
- Used first-principles materials reasoning to troubleshoot coating performance and translate process variables into actionable improvements.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Developed and optimized CVD polymer thin-film deposition with DOE strategies and in-situ QCM-D monitoring to quantify growth kinetics and coating reproducibility.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, safety, and equipment-tracking systems that improved instrument uptime by 30% and strengthened data quality.

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Materials Reliability and Failure-Mode Study Chemistry of Materials 2025

Mapped degradation, redox instability, and performance failure modes across 15+ functional material variants using orthogonal characterization and stress-test logic.

FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC, MS

Bio-Based Polymer Process Qualification Current

Defined DOE-guided process windows, analytical readouts, reproducibility criteria, and SOP-ready material-quality specifications for polymer scale-up.

DOE, SPC, TGA/DSC, spectroscopy, process documentation

AFOSR Defense Thin-Film Coatings DoD Funded

Established process-property relationships for CVD/surface-preparation workflows and reproducible high-performance coating conditions.

CVD, QCM-D, surface prep, thin films

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Materials Testing & Characterization: TGA, DSC, FT-IR, UV-Vis-NIR, microscopy exposure, NMR, GPC, mass-spectrometry-informed analysis, cyclic voltammetry, QCM-D

Reliability & Failure Analysis: Root-cause analysis, degradation studies, anomalous-data interpretation, coatings/thin-film failure modes, material selection recommendations

Laboratory Quality Systems: SOPs, calibration mindset, reproducibility criteria, test-method documentation, data traceability, technical training

Statistical & Process Methods: DOE, SPC, process windows, technical reports, data visualization, Excel, Python/basic data analysis, JMP/basic

Materials Platforms: Polymers, coatings, thin films, organic/organometallic materials, adhesives/composites adjacency, electrochemical materials

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)